

openFDA



FDA has some of the world's most valuable data stores about human health and medicine publicly available on its web site. The data are relevant and important to a wide range of groups. But the hurdles for a developer or researcher to use FDA data can be exceedingly high. For example, someone wanting to use a structured XML document of known adverse event reports for a particular drug would have to go through many complex steps to be able to obtain accurate results.

Now, **openFDA** (<https://open.fda.gov/index.html>), an FDA Health Informatics initiative launched in June 2014, is making it easier for researchers, scientists, web developers, and other FDA regulatory stakeholders to access and use those datasets in an open standard format.

The project aims to create easy access to public data and a new level of openness and accountability; ensure the privacy and security of public FDA data; educate the public; and save lives.

Members of the scientific community can use openFDA to have their applications automatically query the data through application program interfaces (APIs). OpenFDA increases the efficiency and speed of accessing datasets by using cutting-edge, open-source code modules in a cloud-based environment.

Nimble and user-centered product development methodologies are being used to expand and improve the platform and ensure user friendliness of its functions. The current APIs include:

For Human Drugs*

- labeling (more than 67,000 for drugs currently on the market)
- reports on negative side effects (4.9 million since 2003)
- enforcement reports (4,000 records since 2012)

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* over-the-counter and prescription drugs in the United States, including biological therapeutics and generic drugs

For Devices

- classification of 1,700 distinct types of devices organized into 16 medical specialties (6,000 records)
- registration and listing
 - * companies and establishments registrations (24,000 records)
 - * companies product listings (100,000 records)
- premarket approvals (PMAs) and approval supplements (30,000 records since 1977)
- clearances through premarket notifications (510(k)s) and granted de novo requests (141,000 records since 1976)
- recalls (9,500 records since 2002)
- adverse event reports (4.4 million records since 1991)

For Foods

- enforcement reports (8,600 records since 2012)

To date, there have been more than 20 million API calls, half of which were from outside the United States, more than 6,000 registered users, 21,000 connected systems worldwide, and dozens of new software applications that the community has built. Also, FDA scientists developed a statistics application for analyzing drug and device adverse events and drug labeling, using an open statistics package, [Shiny R \(http://shiny.rstudio.com/\)](http://shiny.rstudio.com/)  (<http://www.fda.gov/about-fda/website-policies/website-disclaimer>), that is being added to openFDA.

FDA's Scientific Computing Day Posters

September 9-10, 2019

- [HetEnc: A Deep Learning Predictive Model for Multi-type Biological Dataset Using Neural Network to Identify CDER Manufacturing Sites with Non-compliant Good Manufacturing Practice \(GMP\) Inspection Outcomes \(/media/130492/download?attachment\)](/media/130492/download?attachment)
- [Integrating Adverse Outcome Pathways \(AOPs\) and High Throughput in vitro Assays for Better Risk Evaluations, a Study with Drug Induced Liver Injury \(DILI\) \(/media/130493/download?attachment\)](/media/130493/download?attachment)

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- [Utilizing Neural Network to Identify CDER Manufacturing Sites with Non-compliant Good Manufacturing Practice\(GMP\) Inspection Outcomes \(/media/130494/download?attachment\)](/media/130494/download?attachment)
- [R Shiny Based Mobile Tools To Aid Regulatory Decisions And Program Evaluation \(/media/130495/download?attachment\)](/media/130495/download?attachment)
- [Rapid Geographic Risk Assessment and Model Development Using GREAT and BRisk Applications \(/media/130520/download?attachment\)](/media/130520/download?attachment)
- [HYDRA: An Open-Source Platform for Automated Mining and Visualization of Regulatory Data Using Natural Language Processing and Machine Learning \(/media/130521/download?attachment\)](/media/130521/download?attachment)
- [An NLP-Based Approach for Metadata Extraction and Analysis fo FDA Guidance Documents \(/media/130499/download?attachment\)](/media/130499/download?attachment)
- [Using Machine Learning on ICD-10 data to Enhance an Expert Anaphylaxis Case Definition \(/media/130522/download?attachment\)](/media/130522/download?attachment)
- [Developing Methods for Image Acquisition and Image Analysis for Species-Level Identifying Food Contaminating Beetles \(/media/130502/download?attachment\)](/media/130502/download?attachment)
- [A Data Anomaly Detection Tool for Site Selection at FDA \(/media/130496/download?attachment\)](/media/130496/download?attachment)
- [Application Scaling Techniques on HPC Clusters \(/media/130500/download?attachment\)](/media/130500/download?attachment)
- [Open Format for ION Channel Datasets from Cardiac Electrophysiology In Vitro Assays under CiPA \(/media/130504/download?attachment\)](/media/130504/download?attachment)
- [Data Visualization Tool to Map CDER's Priorities and Deliverables: Role of Public-Private Partnerships \(/media/130497/download?attachment\)](/media/130497/download?attachment)
- [Implementation of Galaxy to expand the use of command-line based bioinformatics analysis of genome-editing experiments \(/media/130503/download?attachment\)](/media/130503/download?attachment)
- [Peridot: A Novel, Prototype, Web-based Toolbox to Conduct Comprehensive Drug Safety and Pharmacovigilance Activities \(/media/130505/download?attachment\)](/media/130505/download?attachment)
- [Bloat-Ectomy: A Method for the Identification and Removal of Duplicate Text in the Bloated Notes of Electronic Health Records and Other Documents \(/media/130501/download?attachment\)](/media/130501/download?attachment)
- [Advancing the Regulatory Science of Omics via Crowdsourcing on the precisionFDA Platform \(/media/130498/download?attachment\)](/media/130498/download?attachment)

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- [FDA's Virtual Assistant: Utilizing Machine Learning for Automated Customer Service \(/media/130558/download?attachment\)](/media/130558/download?attachment)
- [Terminology and Relationships for a Smokeless Tobacco Product Ontology \(/media/130559/download?attachment\)](/media/130559/download?attachment)
- [CTP Integrated Research and Data System \(CIRDS\) \(/media/130560/download?attachment\)](/media/130560/download?attachment)

[More information about the 2019 Scientific Computing Day Symposium \(/news-events/fda-meetings-conferences-and-workshops/2019-scientific-computing-days-symposium-09092019\)](/news-events/fda-meetings-conferences-and-workshops/2019-scientific-computing-days-symposium-09092019)

Resources For You

- [OpenFDA: Innovative Initiative Opens Door to Wealth of FDA's Publicly Available Data \(http://blogs.fda.gov/fdavoiced/index.php/2014/06/openfda-innovative-initiative-opens-door-to-wealth-of-fdas-publicly-available-data/\)](http://blogs.fda.gov/fdavoiced/index.php/2014/06/openfda-innovative-initiative-opens-door-to-wealth-of-fdas-publicly-available-data/)
- [OpenFDA Provides Ready Access to Recall Data \(http://blogs.fda.gov/fdavoiced/index.php/2014/07/openfda-provides-ready-access-to-recall-data/\)](http://blogs.fda.gov/fdavoiced/index.php/2014/07/openfda-provides-ready-access-to-recall-data/) [↗ \(http://www.fda.gov/about-fda/website-policies/website-disclaimer\)](http://www.fda.gov/about-fda/website-policies/website-disclaimer)
- [FDA Leverages Big Data Via Cloud Computing \(http://blogs.fda.gov/fdavoiced/?s=FDA Leverages Big Data Via Cloud Computing&submit=Search\)](http://blogs.fda.gov/fdavoiced/?s=FDA+Leverages+Big+Data+Via+Cloud+Computing&submit=Search)
- [Providing Easy Access to Medical Device Reports Submitted to FDA Since the Early 1990s \(http://blogs.fda.gov/fdavoiced/?s=Providing Easy Access to Medical Device Reports&submit=Search\)](http://blogs.fda.gov/fdavoiced/?s=Providing+Easy+Access+to+Medical+Device+Reports&submit=Search)
- [OpenFDA Makes Medical Device-Related Data Easier to Access and Use \(http://blogs.fda.gov/fdavoiced/?s=OpenFDA Makes Medical Device-Related Data Easier to Access and Use&submit=Search\)](http://blogs.fda.gov/fdavoiced/?s=OpenFDA+Makes+Medical+Device-Related+Data+Easier+to+Access+and+Use&submit=Search)

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